University of Texas at Austin · McCombs School of Business · FIN 285F

Novelty-Driven Mean Reversion

A Portfolio Sort Study of Earnings Conference Call Text Analytics

Zac Garland & Ryota Yokoyama
April 2026

181,305 · S&P 500 universe (Hypotheses Validation) · 2005–2025

Research Question & Hypotheses

Research Question

Do firms whose earnings call transcripts contain semantically novel and financially salient content relative to their own historical baseline generate predictable cross-sectional differences in risk-adjusted returns — and does a long-short portfolio exploiting this signal earn significant alpha?

H1 · Overreaction

Q1 (familiar language) minus Q5 (novel) long-short portfolio earns positive alpha. Benchmark: Cohen, Malloy & Nguyen (2020) 188 bps/month from 10-K language changes.

H2 · Mean Reversion

High-novelty firms (Q5) show larger initial abnormal returns at t=0, followed by partial reversal in post-event window [+2, +5], consistent with overreaction and subsequent correction.

H3 · Sentiment Interaction

Overreaction amplified when novel content carries positive FinBERT tone. Long-short alpha is largest in the double-sorted high-novelty × high-positive-tone portfolio.

Mechanism: Markets overreact to genuinely novel disclosures when sentiment-driven traders and algos lack a reference frame — sophisticated investors then correct the mispricing.

Data Sources & Sample Overview

181,305
Earnings Call Events

6,294
Unique Tickers

15 yrs
Apr 2010 – Dec 2025

8,089
Median Words / Call

Financial Data

Daily Returns
ret_t−15 through ret_t+15 (31-day event window); 3-month holding period for portfolio sort

Overnight Return
close_to_open_return — immediate earnings surprise proxy

Identifiers
CRSP permno for risk adjustment; ~5,093 obs excluded for missing permno

Text & Sample Quality

Transcript Input
full_transcript_text; boilerplate stripped via regex; turn-level embedding (no fixed chunking)

Inclusion Criteria
Event type = Earnings Call · permno non-missing · ≥ 2 prior transcripts · word_count > 100 · full [−1,+5] return window

Outlier Handling
Returns winsorized 1st/99th pct; novelty scores bounded [0,1] by construction

Multi-Layer NLP Pipeline

01

Turn-Level Embedding

nomic-embed-text-v1.5 (768-dim)
Turn = natural utterance unit

02

Novelty Scoring

1 − cos(embed, $centroid_{hist}$)
Rolling 4-quarter or exp-decay baseline

03

Salience Filter

26-cluster weighted dictionary
score ≥ 0.30 required to pass

04

LLM Intent Filter

Ollama local LLM
Binary YES/NO: genuine business catalyst?

05

Macro Penalty

Sigmoid decay at 40% cross-firm frequency; eliminates systemic topics

▶ Output: Adjusted Novelty Score per firm-event

Mean across gate-passing turns × Cross-market frequency penalty → Within-quarter quintile Q1–Q5

LLM Prompt

```
prompt = (
    f'You are a financial analyst reviewing a sentence from an earnings call transcript. Sentence: "{sentence}"\n\n'
    "Does this sentence contain information that is likely to be financially material and meaningfully different from "
    "routine earnings call language? This includes, but is not limited to, new strategic initiatives, product or service "
    "changes, partnerships, contracts, market expansion, pricing decisions, cost actions, management changes, "
    "capital allocation shifts, or any forward-looking statement that represents a departure from prior guidance or "
    "company narrative. Exclude only clear boilerplate (safe-harbour disclaimers, operator instructions, generic "
    "pleasantries) and retrospective descriptions of already-reported results with no forward-looking content. Reply "
    "with ONLY one word: YES or NO."
)
```

Sentence Level Filtering

Ticker : NVDA | Quarter: 2024Q1
Novelty: 0.922 | Salience: 3.880 | Cluster: geographic_expansion
Text : Although we have not received licenses from the U.S. government to ship restricted products to China, we have started shipping alternatives that don't require a license for the China market....

Turn Level Filtering

Ticker : NVDA | Quarter: 2017Q1 | Transcript: 1128909.0
Novelty: 0.893 | Salience: 24.880 | Cluster: technology_digital
Text : Now the reason why deep learning has just swept the world, it started with composition of networks but reinforcement networks and time sequence networks, and all kinds of interesting adversarial networks. The list of types of networks, I mean, there are 100 networks are being created a week. And our papers are coming out ...

▶ Output: Filtered Sentences or Turns for Scoring

Pass Rates (Sentences)/(Turns): (~5-10%)/(~40-50%)

Adjusted Novelty Score Construction

Stage 1 · Precision Gate (ALL three must pass)

✓ **Novelty ≥ 0.60**

Turn semantically distant from firm historical centroid

✓ **Salience ≥ 0.30**

Turn contains financially material content (26-cluster dict)

✓ **LLM = YES**

Ollama classifies turn as genuine business catalyst

Stage 2 · Additive Strength Score (gate-passing turns)

W **Section Weight**

1.0 prepared remarks / 1.5 Q&A (less scripted, analyst-elicited)

A **Analyst Attention**

Fraction of unique analysts probing the topic cluster (revealed-preference filter)

D **Topic Depth**

Turn fraction per cluster, normalized by median, capped at 2.0×

P **Macro Penalty**

Sigmoid decay: f_0=0.40, k=15 — smooth discount for systemic topics

Lexical Baselines (robustness)

Firm-level rolling TF-IDF · 3-month universe TF-IDF · BM25 (k_1=1.5, b=0.75)

```
AdjNovFinal = AdjNov × (⅓ section + ⅓ attention + ⅓ depth) × penalty(f)
```

Top 10 Most Novel + Salient Sentences — Illustrative Examples

Ticker	Quarter	Novelty	Salience	Cluster	Key Insight
NVDA	**2017Q1**	**0.909**	**24.9**	technology	Jensen explains deep learning as software that writes itself from data — GPUs as the enabling infrastructure for modern AI
NVDA	**2011Q1**	**0.876**	**4.4**	guidance	Royalties are an ongoing, innovation-driven revenue line — not one-offs — and will remain core to the business model going forward
MSFT	**2024Q4**	**0.869**	**44.8**	technology	Copilot → Copilot Studio → autonomous agents: full AI system stack described, with Azure as underlying platform for developers to replicate it
AAPL	**2017Q3**	**0.859**	**1.4**	geo expansion	Analyst asks about Trump's 'big beautiful plans' pledge and Apple's working relationship with Chinese government
NVDA	**2015Q3**	**0.845**	**1.4**	product innov	Deep learning + sensor fusion (LiDAR, radar, sonar, camera) described as the only viable path to full autonomous driving
NVDA	**2016Q1**	**0.841**	**22.1**	technology	GPU-accelerated deep learning framed as a new computing paradigm — every hyperscaler and every industry will adopt it
MSFT	**2011Q3**	**0.836**	**2.9**	product innov	Analyst asks how to model Android patent royalty income and which segment it would appear in
NVDA	**2011Q1**	**0.828**	**2.6**	guidance	28nm not available this year; 40nm more economical through mid-next year — production ramp on track with high yields
NVDA	**2023Q2**	**0.823**	**6.0**	technology	Analyst probes inference vs. training split for generative AI — is inference 5x or 10x the opportunity vs. training?
AAPL	**2015Q1**	**0.812**	**0.9**	cost structure	Some FX exposures are unhedgeable (no instruments or prohibitive cost) — full volatility pass-through in those regions

Exploratory Data Analysis · Summary Statistics

Panel A · Financial Variables

Variable	N	Mean	Std	p1	p50	p99
ret_t=0	185,622	0.0004	0.0699	−0.200	0.000	0.196
ret_t=+1	185,628	−0.0010	0.0741	−0.224	0.000	0.217
Close-to-Open	185,588	0.0002	0.0518	−0.176	0.000	0.159

Panel B · Text Variables

Variable	N	Mean	Std	p25	p50	p99
Word Count	202,357	9,624	6,457	5,760	8,089	33,380
Length (chars)	202,357	56,275	37,491	33,893	47,229	194,245

2.1×

Event-day σ vs. non-event days

0.070 vs. ≈ 0.033

−0.045

ret_t−1 / ret_t0 correlation

Mild mean reversion signal

~0 bps

Mean event-day return

Near-zero across full sample

Portfolio Construction Methodology

Portfolio Formation

Ranking Period:

End of calendar quarter t, using AdjNov from current-quarter transcripts

Holding Period:

Quarter t+1 (one quarter); consistent with Cohen et al. scaled to quarterly rebalancing)

Weighting:

Equal-weighted within each quintile; value-weighted versions as robustness checks

Short Leg:

Assumes frictionless shorting; costs acknowledged — high-cost to borrow names hedged via puts or index

Benchmark:

Calendar-time portfolio α vs. Cohen, Malloy & Nguyen (2020) 188 bps/month from 10-K changes

Return Computation

Quintile return: Equal-weighted mean across all firms in quintile q per month

Long-short spread: R_LS = R_Q1 − R_Q5 (buy familiar, short novel)

Risk Adjustment (FF5)

$$R_LS - Rf = \alpha + \beta_1 MKT + \beta_2 SMB + \beta_3 HML + \beta_4 RMW + \beta_5 CMA + \varepsilon$$

α = novelty strategy alpha net of standard risk exposures
Standard errors clustered by calendar date (cross-sectionally correlated events)

Model Specifications

M1

Quintile Sort (main)

$$CAR[-1,+1] = \alpha + \beta_1 Q2 + \beta_2 Q3 + \beta_3 Q4 + \beta_4 Q5 + \gamma SUE + \gamma CtO + \gamma \log(mktcap) + \ldots$$

Non-parametric novelty-return relationship. β_4(Q5) tests H1.

M2

Continuous + Quadratic

$$CAR[-1,+1] = \alpha + \beta_1 NovScore + \beta_2 NovScore^2 + \gamma SUE + \gamma CtO + \gamma \log(mktcap) + \ldots$$

Tests for non-linear (U-shaped or inverted-U) novelty-return relationship.

M3

Mean Reversion (H2 test)

$$CAR[+2,+5] = \alpha + \beta_1 Q5 + \beta_2 CAR[-1,+1] + \beta_3 (Q5 \times CAR[-1,+1]) + \gamma SUE + \ldots$$

$\beta_3 < 0$: high-novelty firms with large initial moves show stronger reversal.

M4

Pre-Priced Novelty

$$CAR[-1,+1] = \alpha + \beta_1 NovScore + \beta_2 |pre_drift| + \beta_3 (NovScore \times |pre_drift|) + \gamma SUE + \ldots$$

$\beta_3 < 0$: novelty effect attenuated when market has already moved pre-call.

Controls across all models: SUE · close_to_open · log_mktcap · book_to_market · analyst_coverage · log_word_count · pre_drift · qa_ratio · Quarter FE · Year FE

Identification Strategy & Key Challenges

Omitted Variable Bias

Novelty may correlate with size, growth, and industry. Mitigated by quarter/year FEs, and direct controls for market cap, book-to-market, and analyst coverage.

Simultaneity

Novelty is measured from the same event generating returns. Rolling historical baseline isolates novelty as deviation from prior-period language — text is produced before market reacts.

Selection Bias

Pipeline scores ALL firms, not just high-novelty ones. Quintile assignment is within-quarter cross-sectional, so even Q1 firms contribute to identification.

Pre-Priced Novelty

Content flagged as novel may already be priced via press releases / 8-Ks. Addressed via: pre_drift control, Model 4 interaction term, topic depth filter, and analyst attention filter.

Cross-Sectional Dependence

Many firms report earnings on the same day, creating correlated residuals. Standard errors clustered by calendar date throughout all regression specifications.

Future Extension: 8-K Match

Direct topic-level match between transcript novelty and prior 8-K filings would sharpen pre-pricing identification. Deferred due to data generation scope for 200k+ firm-events.

Study Summary

① Novel multi-layer NLP pipeline: turn-level semantic novelty (nomic-embed-text-v1.5) + 26-cluster salience filter + local LLM intent gate + sigmoid macro penalty.

② Cross-sectional long-short strategy: Q1 (familiar) minus Q5 (novel), held one quarter, risk-adjusted via Fama-French 5-factor model.

③ Benchmark: Cohen, Malloy & Nguyen (2020) — 188 bps/month alpha from 10-K language changes. We extend to higher-frequency earnings call transcripts.

④ Three hypotheses: Overreaction (H1) · Mean Reversion (H2) · Sentiment Interaction (H3). Four regression models testing each mechanism.

⑤ Dataset: 202,357 earnings calls · 6,867 tickers · April 2005–December 2025 · 185,622 observations with full return windows.